MTH2003

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Differential Equations

Module Leader: Prof. Vadim N. Biktashev

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

USEFUL FORMULAS

These can be used without proof if and where needed.

Expansion in an orthogonal set of functions If $(\phi_m(x))$, $m = 1, 2, \ldots$ is a complete set of functions in a functional space, orthogonal with weight $w(x)$ on an interval $x \in [a, b]$, and $f(x)$ belongs to that space, then

$$f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x), \qquad c_n = \frac{\int_a^b w(x)\phi_n(x)\, f(x)\, \mathrm{d}x}{\int_a^b w(x)\phi_n(x)^2\, \mathrm{d}x}.$$

In particular, the classical Fourier series

$$f(x) = c_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right], \quad x \in [A, A + 2L],$$

$$c_0 = \frac{1}{2L} \int_A^{A+2L} f(x)\, \mathrm{d}x, \quad a_n = \frac{1}{L} \int_A^{A+2L} f(x) \cos\left(\frac{n\pi x}{L}\right)\, \mathrm{d}x,$$

$$b_n = \frac{1}{L} \int_A^{A+2L} f(x) \sin\left(\frac{n\pi x}{L}\right)\, \mathrm{d}x.$$

SECTION A

1. (a) Find any special and typical solutions to

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} = 2y.$$

Further, find and plot the solutions of this equation subject to initial conditions: (A) $y(0) = 1$, (B) $y(1) = 0$, in the maximal intervals where these solutions are unique. For the plots, use ranges $x \in [-2, 1]$, $y \in [-1, 2]$. (15)

(b) Find the general solution $y(x)$ to the following ordinary differential equation:

$$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6x\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 4\,x^2, \qquad x > 0.$$

Hence find the particular solution such that $y(1) = 0$, $y(2) = -2$. (10)

(c) Consider the eigenfunction problem

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0,\ x \in [0,1]; \qquad y(0) = 0,\ y(1) = 0.$$

(i) Bring this equation into the Sturm-Liouville form and hence state the orthogonality condition for the eigenfunctions of this problem.
(ii) Verify that $\phi_n = \mathrm{e}^x \sin(n\pi x)$, $n = 1, 2, 3, \ldots$ are eigenfunctions of this problem and find the corresponding eigenvalues λ_n.
(iii) Verify by direct integration that these eigenfunctions satisfy the orthogonality condition you stated earlier. (15)

(d) Write down the general solution of the partial differential equation with boundary conditions:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial x}, \qquad u(0,t) = 0,\ u(1,t) = 0,$$

in the form of an infinite series of terms of the form $X(x)T(t)$ where $X(x)$ and $T(t)$ should be found (you may, and are indeed encouraged, to use the information given in the question part 1c). Find the coefficients in this series required to satisfy the initial conditions $u(x,0) = f(x)$ and $u_t(x,0) = 0$, in the form of integrals involving $f(x)$. (10)

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SECTION B

2. Consider the system of equations

$$\text{(Eq2.1)} \qquad \begin{aligned} \frac{\mathrm{d}x_1}{\mathrm{d}t} &= -9x_1 + 2x_2 + g_1(t), \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} &= 2x_1 - 6x_2 + g_2(t), \end{aligned}$$

where $g_1(t)$, $g_2(t)$ are given functions of time.

(a) Write system (Eq2.1) in the vector form $\mathrm{d}\mathbf{x}/\mathrm{d}t = \mathbf{A}\mathbf{x}+\mathbf{g}$. Identify $\mathbf{x}$, $\mathbf{A}$ and $\mathbf{g}$. Find the eigenvalues and eigenvectors of the matrix of coefficients $\mathbf{A}$. Hence write down the general solution $x_1(t)$, $x_2(t)$ of (Eq2.1) for the homogeneous case, that is $g_1(t) \equiv 0$, $g_2(t) \equiv 0$. (13)

(b) Find the solution of system (Eq2.1) for $g_1(t) = 5\,\mathrm{e}^{-2t}$, $g_2(t) = 2\,\mathrm{e}^{-2t}$, for the initial conditions $x_1(0) = 1$, $x_2(0) = 1$. (7)

(c) Sketch the phase portrait of the homogeneous system solved in part (a). In the same plot, show also the trajectory corresponding to the solution of the nonhomogeneous initial-value problem found in part (b). (5)

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3. Consider the following ODE:

(Eq3.1) $$y'' - 2xy' + \lambda y = 0.$$

(a) Explain why $x = 0$ is an ordinary point of this equation. Consider solutions in the form of a power series, $y(x) = \sum_{n=0}^{\infty} a_n x^n$, where $a_0, a_1, \ldots$ are constants to be determined, and show that the coefficients a_n satisfy the recurrence relation

$$a_{n+2} = \frac{2n - \lambda}{(n+1)(n+2)} a_n.$$

(7)

(b) Let $y_1(x)$ be the power series solution of equation (Eq3.1) that satisfies initial conditions $y(0) = 1$, $y'(0) = 0$, and $y_2(x)$ be the power series solution that satisfies initial conditions $y(0) = 0$ and $y'(0) = 1$. Write down explicitly the first three nonzero terms in each of these power series. (7)

(c) (i) Find all the values of the parameter λ at which either of the series solutions, y_1 or y_2, terminates and becomes a polynomial. State explicitly, which series will terminate in what case. Write down the resulting polynomial solutions for the differential equation, up to and including the solution which is a cubic polynomial, and the values of λ at which these polynomial solutions exist.
(ii) Let m be the degree of a polynomial solution of (Eq3.1). Verify by substitution that for even m, $m = 2\ell$, the coefficients of the polynomial can be chosen as

$$a_{2k} = \frac{(-1)^k \, 2^{2k}}{(2k)! \; (\ell - k)!}, \qquad a_{2k+1} = 0, \qquad k = 0, \ldots, \ell,$$

and for odd m, $m = 2\ell + 1$, as

$$a_{2k} = 0, \qquad a_{2k+1} = \frac{(-1)^k \, 2^{2k+1}}{(2k+1)! \; (\ell - k)!}, \qquad k = 0, \ldots, \ell.$$

(11)

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4. Consider the two-dimensional Laplace equation in polar coordinates,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}=0, \tag{Eq4.1}$$

in the domain outside the unit circle, $r \geq 1$.

(a) By separating variables, show that $u(r,\theta)$ satisfying equation (Eq4.1) and bounded as $r \to \infty$, is given by

$$u(r,\theta)=A_0+\sum_{n=1}^{\infty} r^{-n}\left(A_n\cos(n\theta)+B_n\sin(n\theta)\right) \tag{Eq4.2}$$

where A_0, A_n, B_n, $n=1,2,\ldots$ are arbitrary constants. (13)

(b) Find the coefficients in the series solutions (Eq4.2) which would satisfy the Dirichlet boundary conditions

$$u(1,\theta)=f(\theta), \tag{Eq4.3}$$

in the form of integrals involving the function $f(\theta)$. (7)

(c) Now find the coefficients in the series solutions (Eq4.2) which would satisfy the Neumann boundary conditions

$$u_r(1,\theta)=g(\theta), \tag{Eq4.4}$$

in the form of integrals involving the function $g(\theta)$. Here, as usual, u_r stands for $\partial u/\partial r$. Does the solution to this boundary-value problem exist for any $g(\theta)$? When the solution exists, is it unique? (5)

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